Geometric Limits of Defense Design: Impossibility Theorems for Prompt Injection Defenses

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Abstract

We characterize the geometric limits of prompt injection defense. We prove three results of increasing strength: (1) any continuous, utility-preserving defense on a connected Hausdorff space must fix boundary points of the alignment deviation surface; (2) when both defense and alignment function are Lipschitz, the defense cannot uniformly reduce alignment deviation far below threshold on a positive-measure band; (3) under a transversality condition, the defense leaves a positive-measure region genuinely unsafe. We also establish parallel discrete results requiring no topology, and extend to multi-turn, stochastic, and pipeline settings. A representation-independent meta-theorem unifies all paths: utility preservation plus bounded regularity forces the defense's fixed-point set to spill beyond the safe region. The full theory is machine-verified in Lean 4 (39 files, ∼300 theorems, zero `sorry`, three standard axioms) and compared with empirical findings from three frontier LLMs [4]. Artifact: `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities?

The defense mechanisms proposed to date—input classifiers [1], output filters [9], constitutional rewriting [3]—share the same structure: a function $D\colon X \to X$ that preprocesses prompts, mapping unsafe inputs to safe equivalents while leaving safe inputs alone. We study the geometric limits of this mapping.

We prove three results of increasing strength. **Tier 1** (Theorem 4.1): if D is continuous, utility-preserving, and the space is connected, D must fix boundary points ($f(z) = \tau$, $D(z) = z$). **Tier 2** (Theorem 5.1): if also Lipschitz, $f(D(x)) \geq \tau - L(K+1)\delta$ for x within distance δ of the fixed point. **Tier 3** (Theorem 6.2): under transversality, the defense leaves positive-measure volume genuinely unsafe. A representation-independent meta-theorem (Theorem 9.1) unifies all paths: utility preservation plus any form of bounded regularity forces $\mathrm{Fix}(D) \supsetneq S_\tau$.

The novelty is not in the proof techniques—which are elementary—but in the problem formulation: identifying that standard defense design constraints induce a forced fixed-point structure at the safety boundary. All results are machine-verified in Lean 4 with zero `sorry` statements.

*Corresponding author: `manish.bhatt13212@gmail.com`. Independent work.

2 Related Work

Adversarial robustness. Prior work characterizes the existence of adversarial perturbations or provides certified robustness within bounded regions [14, 8, 7, 13, 10]. Our result is a structural constraint on the defense mapping itself, independent of perturbation models or training.

Prompt injection. Attack methods discover individual failures [17, 6, 11, 12]. The Manifold of Failure framework [4] maps their topology. We prove these failures are structurally unavoidable under mild constraints.

Impossibility results. No-free-lunch theorems [16] and robustness–accuracy tradeoffs [15] are distributional. Our results hold pointwise for any continuous defense on a fixed task.

Formal verification. Lean and Coq have verified properties of neural networks [2] and optimization [5]. To our knowledge, ours is the first verified impossibility for prompt injection defense.

3 Setup

Definition 3.1. *An* alignment deviation function *is a continuous* $f\colon X \to \mathbb{R}$. *Given threshold* τ*:* $S_\tau = \{f < \tau\}$ *(safe),* $U_\tau = \{f > \tau\}$ *(unsafe),* $B_\tau = \{f = \tau\}$ *(boundary). A* defense $D\colon X \to X$ *is continuous,* utility-preserving *if* $D|_{S_\tau} = \mathrm{id}$*, and* complete *if* $f(D(x)) < \tau$ *for all* x*.*

4 Tier 1: Boundary Fixation

Theorem 4.1 (Boundary Fixation)**.** *Let* X *be a connected Hausdorff space,* f *continuous with* $S_\tau, U_\tau \neq \emptyset$*,* D *continuous with* $D|_{S_\tau} = \mathrm{id}$*. Then* $\exists\, z$ *with* $f(z) = \tau$ *and* $D(z) = z$*. Every* $z \in \overline{S_\tau} \setminus S_\tau$ *satisfies this; this set is nonempty.*

Proof. (1) $\mathrm{Fix}(D)$ is closed (Hausdorff $\Rightarrow$ diagonal closed). (2) $S_\tau \subseteq \mathrm{Fix}(D)$; closed $\Rightarrow \overline{S_\tau} \subseteq \mathrm{Fix}(D)$. (3) S_τ is open, not closed (else clopen, contradicting connectedness). (4) $\overline{S_\tau} \supsetneq S_\tau$; pick z in the difference: $f(z) = \tau$. (5) $z \in \mathrm{Fix}(D)$. □

Corollary 4.2. *No continuous utility-preserving defense is complete.*

Each hypothesis is necessary: disconnected X allows complete defense; discontinuous D allows complete defense; non-utility-preserving D allows complete defense (counterexamples in Section A).

Remark 4.3 (Approximate utility preservation)**.** *If* D *satisfies* $\mathrm{dist}(D(x), x) \leq \varepsilon$ *for all* $x \in S_\tau$ *instead of strict identity, the fixed-point set* $\mathrm{Fix}(D)$ *may not contain* S_τ*. However, the set* $\{x : \mathrm{dist}(D(x), x) \leq \varepsilon\}$ *is closed and contains* S_τ*, so its closure-spillover argument still produces points near the boundary where* D *moves inputs by at most* ε*. The impossibility degrades gracefully: boundary fixation becomes boundary* near*-fixation with displacement* $\leq \varepsilon$*. A defense that "jitters" safe inputs escapes strict fixation but inherits an* ε*-approximate version.*

5 Tier 2: ε-Robust Constraint

Theorem 5.1 (ε-Robust Constraint)**.** *Under Theorem 4.1's hypotheses, if* f *is* L*-Lipschitz and* D *is* K*-Lipschitz, the fixed point* z *satisfies* $f(D(x)) \geq \tau - L(K+1)\,\mathrm{dist}(x, z)$ *for all* x*.*

Theorem 5.2 (Positive-Measure Band)**.** *The ε-band $\{\tau - \varepsilon \le f \le \tau\}$ has positive measure and contains* $\mathrm{ball}(c, \varepsilon/(4L))$ *for c with $f(c) = \tau - \varepsilon/2$.*

Theorem 5.1 constrains the *depth* to which the defense can push near-boundary points below τ. It does not prove they remain unsafe—the defense may map them to $[\tau - L(K+1)\delta, \tau)$.

6 Tier 3: Persistent Unsafe Region

Definition 6.1. *The* steep region $\mathcal{S} = \{x : f(x) > \tau + L(K+1)\,\mathrm{dist}(x, z)\}$.

Theorem 6.2 (Persistent Unsafe Region)**.** *If $\mathcal{S} \neq \emptyset$: (1) $\mathcal{S}$ is open; (2) $\mu(\mathcal{S}) > 0$; (3) $f(D(x)) > \tau$ for all $x \in \mathcal{S}$.*

$\mathcal{S}$ is nonempty whenever the boundary slope exceeds $L(K+1)$—the surface rises faster than the defense can pull it down. Verified in Lean as `transversality_from_deriv`.

7 Discrete-Continuous Bridge and Discrete Path

Theorem 7.1 (Continuous Relaxation)**.** *Finite observations $g(p) < \tau < g(q)$ in a connected normal space admit continuous extensions for which Tiers 1–3 hold (Tietze/McShane–Whitney). This does not claim the results hold for the discrete token system directly, but for every continuous model consistent with the observations.*

Independent of Tietze, we prove discrete results requiring no topology (Section B): a discrete IVT on finite ordered sets, discrete defense fixation, and capacity exhaustion via pigeonhole.

8 Extensions

Multi-turn (Theorem C.1): boundary fixation recurs at every turn; the attacker accumulates information and can steer toward transversality. **Stochastic**: if $g(x) = \mathbb{E}[f(D(x))]$ is continuous and utility-preserving in expectation, $g(z) = \tau$ at boundary points. **Capacity parity**: utility preservation consumes $|S_\tau|$ capacity units, giving the attacker a permanent edge. **Pipelines**: Lipschitz constants multiply through tool calls; n-stage pipeline has effective constant K^n, exponentially degrading the defense.

9 Meta-Theorem

Theorem 9.1 (Regularity Implies Spillover)**.** *If $D(x) = x$ for all $x \in S$ and* $\mathrm{Fix}(D)$ *cannot equal S (bounded regularity), then* $\mathrm{Fix}(D) \supsetneq S$ *and $\exists\, x \notin S$ with $D(x) = x$.*

Each path proves regularity differently (continuous: S not clopen; discrete: IVT + pigeonhole; stochastic: $g - f$ extends to closure). The meta-theorem is representation-independent.

10 Theory–Experiment Correspondence

We compare theoretical predictions with empirical findings from behavioral mapping of three LLMs [4] over a 2D space (query indirection $\times$ authority framing). The comparison is summarized in Table 1.

Llama-3-8B (AD 0.93): Tier 3 territory—slope exceeds $L(K+1)$. **GPT-OSS-20B** (AD 0.73): rugged, many transversal crossings. **GPT-5-Mini** (peak AD 0.50): $U_{0.5} = \emptyset$; no tier applies—strategy is to make boundary harmless.

Table 1: Theoretical predictions and corresponding empirical observations.

Theorem	**Predicts**	**Observed**
Basin (D.1)	Positive-measure basins	Extended heatmap regions
Persistent (6.2)	Steep boundary → unsafe volume	Llama's .93 plateau persists
Stability (E.1)	Deep points survive perturbation	Vulnerability persists across variants
Pipeline (G.1)	Deeper → wider failure band	Tool-using agents harder to defend

11 Engineering Prescription

(1) Make the boundary shallow: set τ so boundary behavior is benign (GPT-5-Mini). (2) Reduce L: smoother surfaces raise the Tier 3 threshold. (3) Reduce dimension: defense cost $= N^d$. (4) Monitor the boundary: crossings persist under fine-tuning. (5) Multi-turn context doesn't help. (6) Randomization fails in expectation. (7) Break capacity parity via architectural separation.

12 Limitations

Tier 1 applies only at the boundary ($f(z) = \tau$). Tier 2 constrains depth, not direction. Tier 3 requires transversality. The Tietze bridge applies to interpolants, not the discrete system itself; positive-measure bands may be empty of valid token sequences (the sharpest open gap). Raising τ fails for binary safety constraints. See Section I for full discussion.

13 Conclusion

We establish geometric limits on prompt injection defense via a three-tier impossibility hierarchy, unified by a representation-independent meta-theorem. The impossibility is a consequence of combining utility preservation with bounded regularity—not an artifact of any particular proof technique. The practical prescription: make the boundary shallow, smooth, and low-dimensional. Machine-verified in Lean 4: 39 files, zero `sorry`.

References

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A Counterexamples

Counterexample A.1 (Removing connectedness). *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample A.2 (Removing continuity). *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$.*

Counterexample A.3 (Removing utility preservation). *$D(x) = x_0$ for a fixed safe point.*

B Discrete Impossibility

Theorem B.1 (Discrete IVT). *$f\colon \{0, \ldots, n+1\} \to \mathbb{R}$ with $f(0) < \tau \leq f(n+1)$: $\exists\, i$ with $f(i) < \tau$ and $f(i+1) \geq \tau$.*

Theorem B.2 (Discrete Defense Fixation). *If D fixes safe vertices, the crossing has its safe endpoint fixed.*

Theorem B.3 (Capacity Exhaustion). *m configs, $< m$ defense behaviors $\Rightarrow$ pigeonhole collision.*

C Multi-Turn Details

Theorem C.1 (Multi-Turn Impossibility). *$\{f_t, D_t\}_{t=1}^{T}$ with each f_t continuous, mixed regions, D_t utility-preserving: $\forall\, t$, $\exists\, z_t$ with $f_t(z_t) = \tau$, $D_t(z_t) = z_t$.*

The attacker's best score is monotone non-decreasing. It can steer toward transversality via binary search on its influence parameter.

D Vulnerability Landscape

Theorem D.1 (Basin Structure). *U_τ is open with positive measure (under any measure positive on nonempty open sets).*

Theorem D.2 (Fragment Size). *L-Lipschitz f, $f(p) > \tau$: component diameter $\geq 2(f(p) - \tau)/L$.*

Theorem D.3 (Perturbation Robustness). $\mathrm{ball}(p, (f(p) - \tau)/L) \subseteq U_\tau$.

Theorem D.4 (Convergence). *Monotone operators on $[0, 1]$ converge in $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem D.5 (Transferability). $\|f - g\|_\infty \leq \delta \Rightarrow \{f > \tau + \delta\} \subseteq \{g > \tau\}$.

Theorem D.6 (Authority Monotonicity). *Monotone in authority $\Rightarrow$ vulnerability upward-closed; threshold curve non-increasing if also monotone in indirection.*

Theorem D.7 (Gradient Ascent). *Nonzero Fréchet derivative $\Rightarrow \exists$ ascent direction.*

E Stability Under Fine-Tuning

Theorem E.1 (Interior Stability). *$\|f - g\|_\infty \leq \varepsilon$: deep basin/safe points preserve classification.*

Theorem E.2 (Crossing Preservation). *Transversal crossings persist under perturbation.*

Theorem E.3 (Patching Is Nonlocal). *Eliminating a vulnerability at one point changes values elsewhere. (Lean construction.)*

F Cost Asymmetry

Theorem F.1 (Grid-Based Cost). *Grid defense: N^d (exponential). Attack: $\leq 1/\delta$ (dimension-free).*

G Pipeline Details

Theorem G.1 (Pipeline Degradation). *n stages compose to $(K_1 \cdots K_n)$-Lipschitz. Failure band scales as $L(K^n + 1)\delta$.*

H Lean Artifact

Verified in Lean 4.28.0 + Mathlib v4.28.0. 39 files: 10 core (MoF_01–10), 10 cost, 10 advanced, MoF_ContinuousRelaxation, MoF_11_EpsilonRobust, MoF_12_Discrete, MoF_13_MultiTurn, MoF_14_MetaTheorem, MoF_15_NonlinearAgents, 3 capstone. Three axioms: `propext`, `Classical.choice`, `Quot.sound`. Zero additional.

I Extended Limitations

Jitter defense. $D(x) \approx x$ breaks strict fixation but inherits ε-approximate version (Theorem 4.3).

Binary exploits. Theorem applies to smooth behavioral landscape, not discrete exploit chains.

Measure gap. Positive-measure bands may be empty of valid tokens. Sharpest open gap.

Non-Hausdorff agents. Geometric lock needs connectedness, not Hausdorff; discrete path needs no topology.

Monitoring blind spot. Shallow boundary may allow tunneling into deep unsafe basins.

Binary safety constraints. "Raise τ" fails for cliff transitions.